

Here's a **clear, instructor-style walkthrough of Git Installation & Setup**—from zero to ready-to-use.

1. What is Git?

Git is a **distributed version control system (DVCS)** used to track changes in source code and collaborate with teams.

It allows you to:

- Track file history
 - Work in branches
 - Collaborate via platforms like GitHub or GitLab
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2. Install Git on Different OS

Windows Installation

Steps:

1. Go to  <https://git-scm.com>
2. Download for Windows
3. Run .exe installer
4. Keep default settings (recommended)
5. Install

Verify:

`git --version`

⚙️ 3. Initial Git Setup (VERY IMPORTANT)

After installation, configure Git identity.

◆ Set Username

```
git config --global user.name "Your Name"
```

◆ Set Email

```
git config --global user.email "your@email.com"
```

👉 This should match your account on GitHub

◆ Check Configuration

```
git config --list
```

🔒 5. Authentication Setup (Important for GitHub)

Option 1: HTTPS (Simple)

Use username + Personal Access Token (not password)

Option 2: SSH (Recommended for professionals)

Generate SSH Key

```
ssh-keygen -t ed25519 -C "your@email.com"
```

Start SSH Agent

```
eval "$(ssh-agent -s)"
```

```
ssh-add ~/.ssh/id_ed25519
```

Add key to:

👉 GitHub → Settings → SSH Keys

📁 6. Initialize Your First Repository

```
mkdir my-project
```

```
cd my-project
```

```
git init
```

👉 This creates a hidden .git folder (Git tracking starts)

+ 7. First Commit (Basic Workflow)

```
git add .
```

```
git commit -m "Initial commit"
```

🔗 8. Connect to Remote Repository (GitHub)

```
git remote add origin https://github.com/username/repo.git
```

```
git push -u origin main
```

📌 9. Verify Everything Works

```
git status
```

```
git log
```

⚡ Summary Flow

Install Git

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Configure (name, email)

↓

Create Repo (git init)



Add Files (git add)



Commit (git commit)



Push to GitHub
